

1. Introduction to ANSYS Workbench

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Aim

To introduce ANSYS Workbench interface and perform a basic static structural analysis of a cantilever beam.

Software Required

ANSYS Workbench (Student / Academic Version)

Problem Statement

Analyse the deformation and stress distribution of a cantilever beam subjected to an external load using ANSYS Workbench.

Theory (Brief)

Static structural analysis is used to determine stresses, strains and deformation caused by applied loads that do not vary with time.

Procedure

1. Open ANSYS Workbench.
2. Drag and drop Static Structural system into the Project Schematic.
3. Double click Geometry and create a rectangular beam using SpaceClaim.
4. Set beam dimensions: Length = 200 mm, Width = 20 mm, Height = 20 mm.
5. Close and save geometry.
6. Open Model and assign Structural Steel as material.
7. Generate mesh using default settings.
8. Apply Fixed Support at one end of the beam.
9. Apply Force of 1000 N downward at the free end.
10. Solve the model.
11. Insert Total Deformation and Equivalent (von-Mises) Stress and note down the results.
12. Change Force to 2000 N, resolve and measure maximum deformation and von-Mises Stresses.

Results

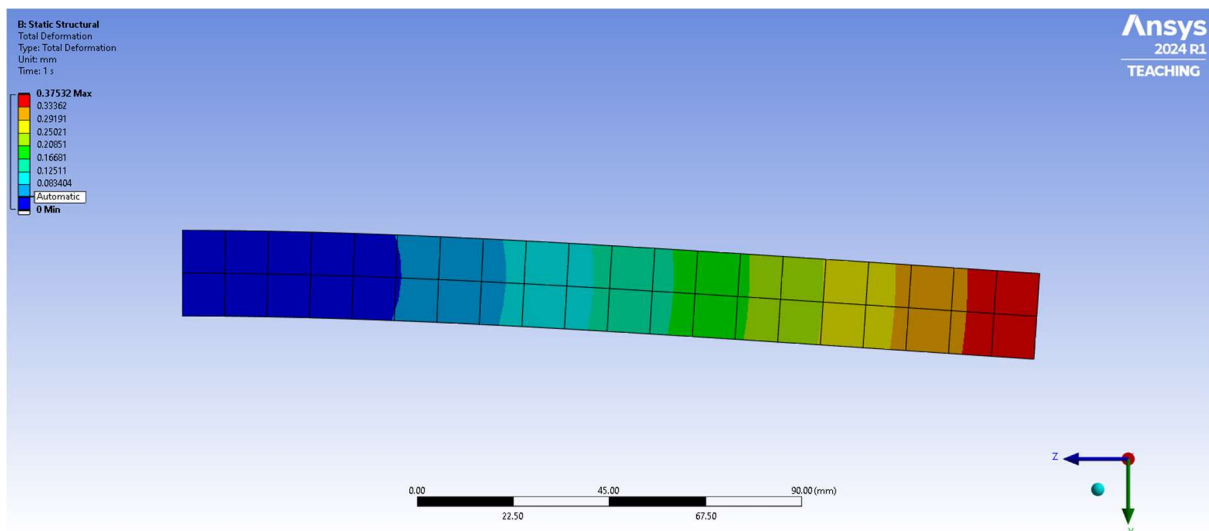
The maximum deformation occurs at the free end of the beam, and maximum stress occurs near the fixed support.

For 1000 N

Total deformation

The value of maximum deformation is 0.37532 mm

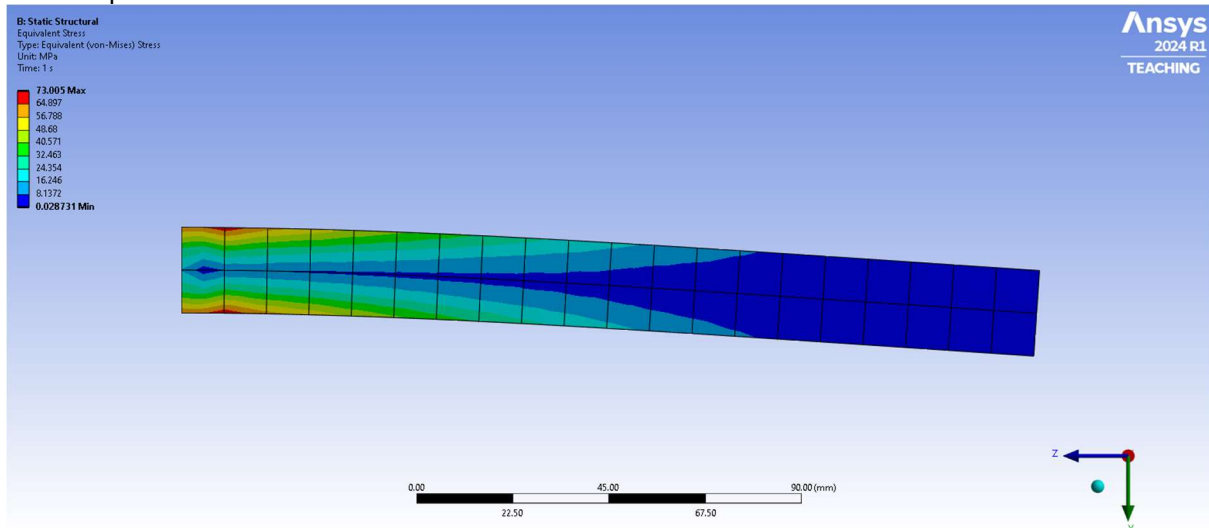
Contour plot:



von-Mises Stress

The value of maximum stress is 73.005 MPa.

Contour plot

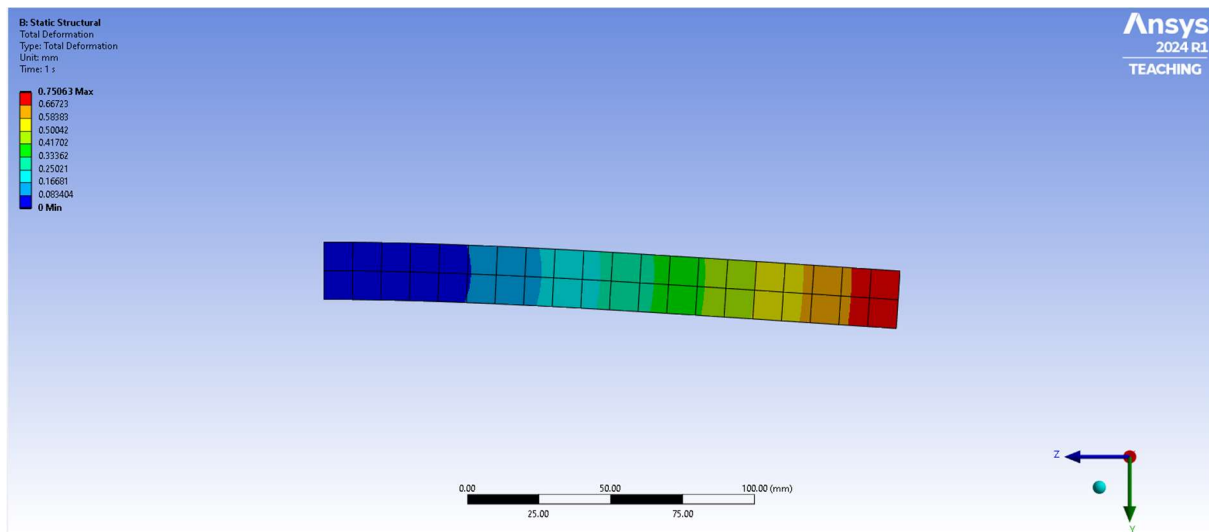


For 2000 N

Total deformation

The value of maximum deformation is 0.75063 mm

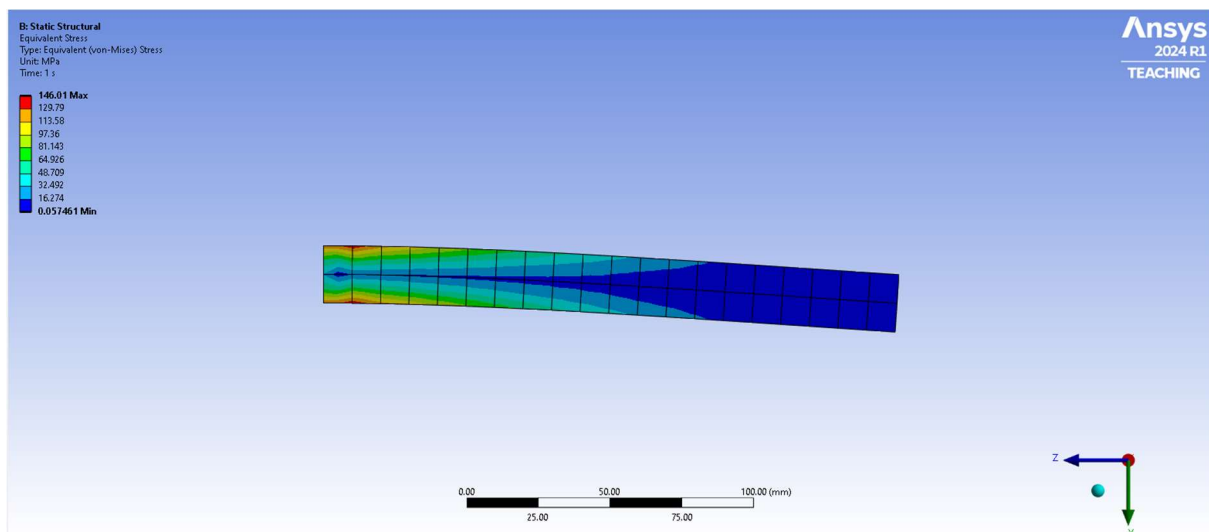
Contour plot:



von-Mises Stress

The value of maximum stress is 146.01 MPa

Contour plot:



Conclusion

ANSYS Workbench was successfully used to perform static structural analysis and deformation and stress distribution are visualized.